

Chapter 4: Well Hydraulics

Comprehensive Notes & Derivations

Syllabus Coverage: Steady/Unsteady Radial Flow, Penetrating Wells, Multiple Well Systems

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1 Introduction to Well Hydraulics

Well hydraulics is a branch of hydrogeology that deals with the theory of flow of fluids (primarily water) into or out of wells. It provides the mathematical framework to determine:

1. The hydraulic properties of aquifers (Transmissivity, Storage Coefficient, Hydraulic Conductivity).
2. The performance and efficiency of wells (Specific Capacity, Well Loss).
3. The effects of pumping on the local water table or piezometric surface (Drawdown, Interference).

Understanding these concepts is critical for the sustainable management of groundwater resources, designing well fields for municipal water supply, and dewatering construction sites.

2 4.1 Definitions & Concepts

These terms describe the physical changes that occur in an aquifer when water is pumped from a well.

2.1 Cone of Depression

When a well is pumped, water is removed from the aquifer surrounding the well. This removal creates a difference in head between the water inside the well and the water in the aquifer.

Definition

The **Cone of Depression** is the conically shaped depression in the water table (unconfined aquifer) or potentiometric surface (confined aquifer) caused by pumping. It is centered at the pumping well and expands outward.

Physical Mechanism

Water flows from high energy (high head) to low energy (low head). Pumping lowers the water level in the well, creating a low-head zone. The surrounding water flows radially toward this low point. The friction of water moving through the soil pores causes energy loss, which results in the curved shape of the water surface.

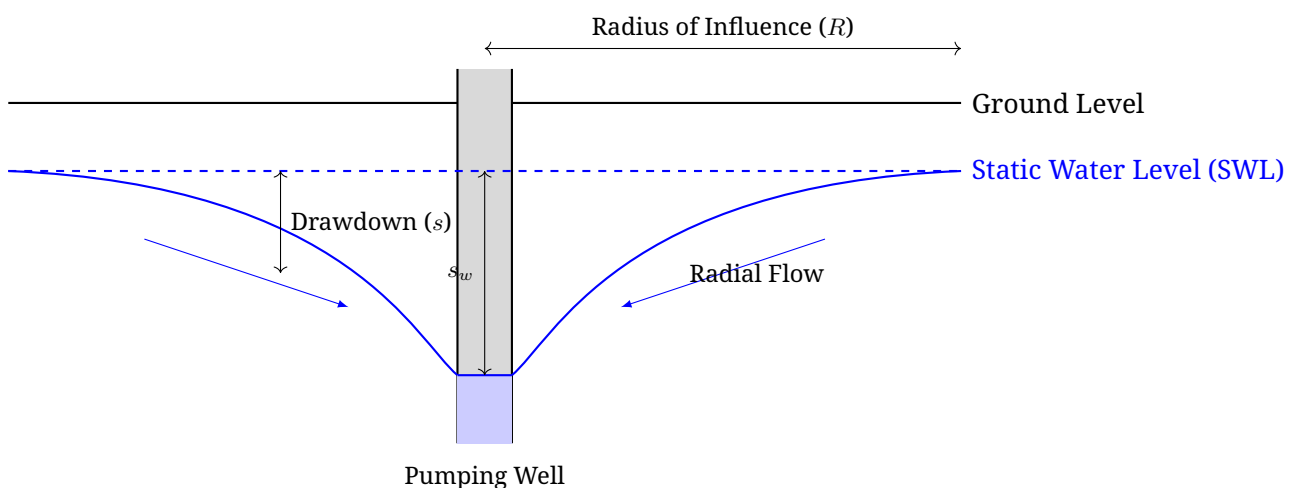


Figure 1: Schematic of the Cone of Depression

The shape of the cone is determined by the aquifer properties:

- **High Transmissivity (T):** The cone is shallow and flat because water moves easily to replenish the well.
- **Low Transmissivity (T):** The cone is steep and deep because high gradients are needed to force water through the formation.

2.2 Drawdown (s)

Definition

Drawdown is the vertical distance between the static water level (before pumping) and the pumping water level at a specific location and time.

Formula

$$s(r, t) = h_0 - h(r, t)$$

Where:

- h_0 = Initial static head
- $h(r, t)$ = Head at distance r and time t

Drawdown is maximum at the well face (s_w) and effectively zero at the radius of influence. In unsteady flow, drawdown increases with time.

2.3 Radius of Influence (R)

Definition

The **Radius of Influence** is the horizontal distance from the center of the well to the limit of the cone of depression where drawdown is negligible.

Empirical Formulas for R

While R theoretically extends to infinity in an infinite aquifer, practical formulas exist for estimation:

1. Sichardt's Formula:

$$R = 3000s_w\sqrt{K}$$

Where:

- R is in meters
- s_w is drawdown in the well (meters)
- K is hydraulic conductivity (m/sec)

2. Cooper-Jacob Approximation (for unsteady flow):

$$R = \sqrt{\frac{2.25Tt}{S}}$$

This shows that R expands as time (t) increases.

2.4 Well Yield (Q)

The **Yield** is the volume of water discharged per unit time (L/s or m^3/day).

- **Safe Yield:** The amount of water that can be withdrawn annually without producing an undesired result (e.g., land subsidence, salt water intrusion).
- **Maximum Yield:** Limited by the physical construction of the well and the aquifer mechanics.

2.5 Specific Capacity (S_c)

Definition

Specific Capacity is the discharge per unit of drawdown in the pumping well. It is a measure of the productivity of the well.

Formula

$$S_c = \frac{Q}{s_w}$$

Factors Affecting Specific Capacity:

1. **Time:** S_c decreases with time as the cone of depression expands and boundaries are reached.
2. **Discharge Rate:** At higher Q , turbulent losses increase, causing disproportionate drawdown, thus lowering S_c .
3. **Well Construction:** Poor screen design or clogging reduces S_c .

3 4.2 Assumptions of Well Hydraulics

To solve the complex differential equations of groundwater flow, simplifying assumptions are necessary.

3.1 General Assumptions (Thiem's Assumptions)

These assumptions apply to Steady State Radial Flow derivation.

1. **Infinite Areal Extent:** The aquifer extends infinitely in all radial directions. This means no boundary conditions (like a river or rock barrier) are encountered within the cone of depression.
2. **Homogeneous & Isotropic:**
 - *Homogeneous:* Properties (K, S) are the same at all locations.
 - *Isotropic:* Hydraulic conductivity is the same in all directions ($K_x = K_y = K_z$).
3. **Uniform Thickness:** The aquifer thickness (b) does not vary.
4. **Steady State:** The flow rate Q is constant, and the water levels have stabilized ($\partial h / \partial t = 0$).
5. **Fully Penetrating Well:** The well is screened across the entire saturated thickness. This ensures flow is strictly horizontal.
6. **Laminar Flow:** Darcy's Law is valid. Reynolds number is low.
7. **Horizontal Flow:** Vertical flow components are negligible.

3.2 Dupuit's Assumptions (For Unconfined Aquifers)

Jules Dupuit (1863) simplified the difficult problem of unconfined flow where the flow thickness decreases near the well.

1. **Horizontal Flow Assumption:** Although the water table is curved, the streamlines are assumed to be horizontal. This means equipotential lines are vertical.
2. **Hydraulic Gradient Assumption:** The hydraulic gradient is constant with depth and is equal to the slope of the water table at the surface.

$$i = \frac{dh}{dr}$$

Implication: These assumptions lead to negligible errors at distances far from the well but may underestimate seepage near the well face where the curvature of the water table is steep.

4 4.3 Steady Radial Flow in Confined Aquifers

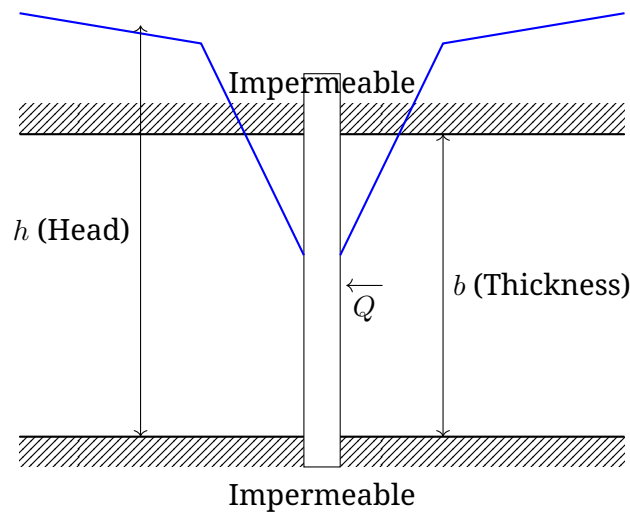
4.1 Concept of Confined Flow

In a confined aquifer, water is bounded between two impermeable layers (aquicludes). The water is under pressure. When pumped:

- The aquifer remains saturated (b is constant).
- The pressure head drops, but the physical pore space is not drained.

4.2 Derivation of Thiem's Equation (Confined)

Consider a fully penetrating well in a confined aquifer of thickness b .



Step 1: Darcy's Law for Radial Flow

$$Q = K \cdot A \cdot i$$

Step 2: Define Area and Gradient At a radial distance r :

- The flow cross-section is a cylinder surface: $A = 2\pi rb$
- The hydraulic gradient is the slope of the piezometric surface: $i = \frac{dh}{dr}$

Step 3: Differential Equation Substitute A and i into Darcy's Law:

$$Q = K(2\pi rb) \frac{dh}{dr}$$

Step 4: Separation of Variables Group r terms on the left and h terms on the right:

$$\frac{dr}{r} = \frac{2\pi Kb}{Q} dh$$

Since $T = Kb$ (Transmissivity):

$$\frac{dr}{r} = \frac{2\pi T}{Q} dh$$

Step 5: Integration Integrate between two observation wells:

- Well 1 at distance r_1 with head h_1

- Well 2 at distance r_2 with head h_2 (where $r_2 > r_1$)

$$\int_{r_1}^{r_2} \frac{1}{r} dr = \frac{2\pi T}{Q} \int_{h_1}^{h_2} dh$$

$$[\ln r]_{r_1}^{r_2} = \frac{2\pi T}{Q} [h]_{h_1}^{h_2}$$

$$\ln r_2 - \ln r_1 = \frac{2\pi T}{Q} (h_2 - h_1)$$

$$\ln \left(\frac{r_2}{r_1} \right) = \frac{2\pi T}{Q} (h_2 - h_1)$$

Step 6: Final Equation Solving for discharge Q :

Thiem's Equation (Confined)

$$Q = \frac{2\pi T(h_2 - h_1)}{\ln(r_2/r_1)}$$

In terms of Drawdown ($s = H - h$): Since $h_2 - h_1 = (H - s_2) - (H - s_1) = s_1 - s_2$:

$$Q = \frac{2\pi T(s_1 - s_2)}{\ln(r_2/r_1)}$$

4.3 Numerical Application (Confined)

Numerical Example: Finding Transmissivity

A pumping test was carried out in a confined aquifer.

- Discharge (Q) = $2500 \text{ m}^3/\text{day}$
- Observation well 1: $r_1 = 15 \text{ m}$, drawdown $s_1 = 3.5 \text{ m}$
- Observation well 2: $r_2 = 45 \text{ m}$, drawdown $s_2 = 1.2 \text{ m}$

Find: The Transmissivity (T) of the aquifer.

Solution: Using Thiem's equation for drawdown difference:

$$T = \frac{Q \ln(r_2/r_1)}{2\pi(s_1 - s_2)}$$

1. Calculate logarithmic term:

$$\ln(45/15) = \ln(3) \approx 1.0986$$

2. Calculate drawdown difference:

$$s_1 - s_2 = 3.5 - 1.2 = 2.3 \text{ m}$$

3. Substitute values:

$$T = \frac{2500 \times 1.0986}{2 \times 3.1416 \times 2.3}$$

$$T = \frac{2746.5}{14.45}$$

$$T \approx 190.07 \text{ m}^2/\text{day}$$

Result: The Transmissivity is approx $190 \text{ m}^2/\text{day}$.

5 4.4 Steady Radial Flow in Unconfined Aquifers

5.1 Concept of Unconfined Flow

In an unconfined aquifer, the upper boundary is the water table itself.

- As water is pumped, the saturated thickness decreases from H to h .
- The flow area A is a function of h , making the integration slightly different.
- Dupuit's assumptions are used.

5.2 Derivation of Dupuit's Equation

Consider a well in an unconfined aquifer overlying a horizontal impervious base.

Step 1: Darcy's Law

$$Q = K \cdot A \cdot i$$

Step 2: Geometry At distance r :

- Flow Area $A = 2\pi r h$ (Note: h is variable)
- Gradient $i = \frac{dh}{dr}$

Step 3: Differential Equation

$$Q = K(2\pi r h) \frac{dh}{dr}$$

Step 4: Separation of Variables

$$\frac{dr}{r} = \frac{2\pi K}{Q} h dh$$

Step 5: Integration Integrate between limits (r_1, h_1) and (r_2, h_2) :

$$\int_{r_1}^{r_2} \frac{1}{r} dr = \frac{2\pi K}{Q} \int_{h_1}^{h_2} h dh$$

$$\ln\left(\frac{r_2}{r_1}\right) = \frac{2\pi K}{Q} \left[\frac{h^2}{2}\right]_{h_1}^{h_2}$$

$$\ln\left(\frac{r_2}{r_1}\right) = \frac{\pi K}{Q} (h_2^2 - h_1^2)$$

Step 6: Final Equation

Dupuit's Equation (Unconfined)

$$Q = \frac{\pi K (h_2^2 - h_1^2)}{\ln(r_2/r_1)}$$

Alternative Form (Using Transmissivity): If the drawdown is small compared to total thickness ($s \ll H$), we can approximate $h_1 + h_2 \approx 2H$. Since $(h_2^2 - h_1^2) = (h_2 - h_1)(h_2 + h_1)$,

$$Q \approx \frac{\pi K (2H)(h_2 - h_1)}{\ln(r_2/r_1)} = \frac{2\pi (KH)(s_1 - s_2)}{\ln(r_2/r_1)}$$

This reduces to the Thiem equation for confined aquifers where $T = KH$.

5.3 Equation of Phreatic Line

To find the height of the water table h_x at any distance r_x : Using the well face (r_w, h_w) as the reference:

$$Q = \frac{\pi K (h_x^2 - h_w^2)}{\ln(r_x/r_w)}$$

Rearranging:

$$h_x = \sqrt{h_w^2 + \frac{Q \ln(r_x/r_w)}{\pi K}}$$

5.4 Numerical Application (Unconfined)

Numerical Example: Unconfined Discharge Calculation

A well fully penetrates an unconfined aquifer.

- Radius of well (r_w) = 0.3 m
- Hydraulic Conductivity (K) = 25 m/day
- Initial saturated thickness (H) = 30 m
- Drawdown at the well (s_w) = 5 m
- Radius of Influence (R) = 300 m

Calculate: The steady discharge (Q).

Solution: 1. Determine heads: At well ($r_1 = r_w$): $h_w = H - s_w = 30 - 5 = 25$ m. At radius of influence ($r_2 = R$): $h_2 = H = 30$ m.

2. Apply Dupuit's Formula:

$$Q = \frac{\pi K (H^2 - h_w^2)}{\ln(R/r_w)}$$

3. Substitute values: Numerator:

$$\begin{aligned} & \pi \times 25 \times (30^2 - 25^2) \\ &= 78.54 \times (900 - 625) \\ &= 78.54 \times 275 = 21598.5 \end{aligned}$$

Denominator:

$$\ln(300/0.3) = \ln(1000) \approx 6.908$$

4. Calculate Q:

$$Q = \frac{21598.5}{6.908} \approx 3126.6 \text{ m}^3/\text{day}$$

Answer: The discharge is 3127 m³/day.

Past Exam Question (2072 Magh)

Derive the equation $Q = \frac{\pi K (h_2^2 - h_1^2)}{\ln(r_2/r_1)}$ for the steady radial flow for unconfined aquifer. [6 Marks]

6 4.5 Unsteady Radial Flow

Steady state conditions are rarely achieved in short-term pumping tests. More commonly, the cone of depression is expanding, and water is being derived from storage.

6.1 Source of Water in Unsteady Flow

- **Confined Aquifer:** Water comes from the compression of the aquifer skeleton and expansion of water (Elastic Storage).
- **Unconfined Aquifer:** Water comes from the actual dewatering of pores (Specific Yield).

6.2 The Theis Equation

Charles V. Theis (1935) developed the non-equilibrium equation based on the analogy between groundwater flow and heat conduction.

Theis Equation

$$s = \frac{Q}{4\pi T} W(u)$$

Where u is a dimensionless parameter:

$$u = \frac{r^2 S}{4Tt}$$

Variables:

- s = Drawdown (m)
- Q = Constant pumping rate (m^3/day)
- T = Transmissivity (m^2/day)
- S = Storage Coefficient (dimensionless)
- t = Time since pumping started (days)
- r = Distance from pumping well (m)
- $W(u)$ = The **Well Function**

The Well Function $W(u)$

$W(u)$ is an infinite series integral (exponential integral):

$$W(u) = \int_u^\infty \frac{e^{-x}}{x} dx = -0.5772 - \ln u + u - \frac{u^2}{2 \cdot 2!} + \frac{u^3}{3 \cdot 3!} - \dots$$

Usage: Values of $W(u)$ are typically found in look-up tables based on the calculated value of u .

6.3 Method of Solution (Theis Type Curve)

To find S and T from pumping test data:

1. Plot the theoretical "Type Curve" ($W(u)$ vs. $1/u$) on log-log paper.
2. Plot field data (s vs. t/r^2) on transparent log-log paper of the same scale.
3. Superimpose the field plot over the Type Curve to find a "Match Point".
4. Use the coordinates of the match point to solve the Theis equation for T and S .

6.4 Cooper-Jacob Approximation

Theis's method is cumbersome due to curve matching. Cooper and Jacob (1946) noted that for **small values of u** ($u < 0.01$), the higher-order terms in the infinite series become negligible.

The series becomes:

$$W(u) \approx -0.5772 - \ln u$$

Converting natural log to base 10 ($\ln u \approx 2.303 \log_{10} u$):

$$s = \frac{Q}{4\pi T} (-0.5772 - \ln \frac{r^2 S}{4Tt})$$

Simplifying algebraically results in:

Cooper-Jacob Equation

$$s = \frac{2.303Q}{4\pi T} \log_{10} \left(\frac{2.25Tt}{r^2 S} \right)$$

Time-Drawdown Method (Calculation Steps)

This method is widely used because it allows graphical solution on semi-log paper.

1. **Plot Data:** Plot Drawdown (s) on the arithmetic y-axis vs. Time (t) on the logarithmic x-axis. 2. **Fit Line:** Draw a straight line through the late-time data points (where u is small). 3. **Find Δs :** Determine the drawdown difference over one log cycle of time (e.g., between $t = 10$ min and $t = 100$ min). This value is Δs . 4. **Calculate T:**

$$T = \frac{2.303Q}{4\pi \Delta s}$$

5. **Find t_0 :** Extend the straight line to where drawdown $s = 0$. Read this time intercept as t_0 . 6. **Calculate S:**

$$S = \frac{2.25Tt_0}{r^2}$$

Numerical Example: Cooper-Jacob Analysis

Given: $Q = 1000 \text{ m}^3/\text{day}$, $r = 20 \text{ m}$. From semi-log plot: $\Delta s = 2 \text{ m}$ (per log cycle), and $t_0 = 1.5$ minutes.

Calculate T:

$$T = \frac{2.303 \times 1000}{4 \times \pi \times 2} = \frac{2303}{25.13} = 91.6 \text{ m}^2/\text{day}$$

Calculate S: Convert t_0 to days: $1.5 \text{ min} = 1.5/1440 \text{ days} = 0.00104 \text{ days}$.

$$S = \frac{2.25 \times 91.6 \times 0.00104}{20^2}$$

$$S = \frac{0.214}{400} = 0.000535$$

(Dimensionless, indicative of a confined aquifer).

7 4.6 Recovery of a Well

When pumping stops, the water level begins to rise (recover). The recovery phase provides an excellent check on the aquifer properties calculated during pumping.

7.1 Theis Recovery Method

Recovery is treated mathematically as injecting water at rate Q into the well (superposition of $+Q$ pumping and $-Q$ recovery).

Residual Drawdown (s'):

$$s' = \frac{2.303Q}{4\pi T} \log_{10} \left(\frac{t}{t'} \right)$$

Where:

- t = Time since pumping started
- t' = Time since pumping stopped (recovery time)

Procedure: Plot Residual Drawdown (s') vs. the ratio t/t' on semi-log paper. The slope allows calculation of T . Storage coefficient S cannot be reliably determined from recovery tests alone.

8 4.7 Partially Penetrating Wells

In many cases, a well is not screened through the entire thickness of the aquifer.

- **Reason:** To save cost, or to avoid poor quality water at the bottom of the aquifer.
- **Effect:** Flow lines must converge vertically to enter the screen. This convergence creates vertical flow components.
- **Impact:** The path of water is longer, increasing friction. This results in **additional draw-down** compared to a fully penetrating well.

$$s_{observed} = s_{fully} + \Delta s_{partial}$$

The effect of partial penetration is negligible at distances $r > 1.5b$ to $2b$. Therefore, observation wells placed far from the pumping well can still use standard Theis/Thiem equations.

9 4.8 Well Loss and Specific Capacity

The drawdown measured inside the pumping well (s_w) is always greater than the theoretical drawdown calculated by Thiem or Theis.

9.1 Components of Drawdown

$$s_w = s_{aquifer} + s_{well}$$

1. **Aquifer Loss (BQ):** The head loss due to laminar flow in the aquifer formation. It is proportional to discharge Q .
2. **Well Loss (CQ^2):** The head loss due to turbulent flow through the well screen and inside the casing. It is proportional to Q^2 .

Jacob's Well Loss Equation

$$s_w = BQ + CQ^2$$

- B = Formation loss coefficient (day/m^2)
- C = Well loss coefficient (day^2/m^5)

9.2 Well Efficiency (E_w)

Well efficiency is the ratio of the theoretical drawdown (aquifer loss only) to the actual measured drawdown.

$$E_w = \frac{BQ}{BQ + CQ^2} \times 100\%$$

- A well is considered efficient if $E_w > 80\%$.
- As Q increases, the CQ^2 term grows rapidly, reducing efficiency.

10 4.9 Multiple Well Systems & Interference

10.1 Principle of Superposition

When multiple wells are pumped in the same aquifer, the drawdown at any point is the algebraic sum of the drawdowns created by each well individually.

$$s_{total} = s_1 + s_2 + s_3 + \dots$$

For steady flow (Confined):

$$s_{total} = \sum_{i=1}^n \frac{Q_i}{2\pi T} \ln \left(\frac{R_i}{r_i} \right)$$

10.2 Example: Three Well System

Consider three wells (W_1, W_2, W_3) forming an equilateral triangle of side L . If all three pump at rate Q , the drawdown at the center of the triangle is the sum of drawdowns from all three.

Distance from each well to centroid = r .

$$s_{center} = s_{W1} + s_{W2} + s_{W3}$$

$$s_{center} = 3 \times \frac{Q}{2\pi T} \ln \left(\frac{R}{r} \right)$$

10.3 Interference

Interference occurs when the cones of depression overlap.

- **Effect:** The water level drops lower than if wells were isolated.
- **Cost:** Pumping lifts increase, requiring more energy ($Power \propto Q \cdot s_{total}$).
- **Management:** Proper well spacing is required to minimize interference.

10.4 Boundary Effects (Image Well Theory)

Aquifers are not infinite. They hit boundaries.

1. **Recharge Boundary (River):** Prevents drawdown from expanding. Modeled by placing an **Image Injection Well** (adding water) equidistant on the other side of the boundary.
2. **Barrier Boundary (Rock):** Prevents flow. Modeled by placing an **Image Pumping Well** (removing water) equidistant on the other side.

11 Appendix: Summary of Formulas

Condition	Formula	Remarks
Steady, Confined	$Q = \frac{2\pi T(s_1 - s_2)}{\ln(r_2/r_1)}$	Thiem's Eq.
Steady, Unconfined	$Q = \frac{\pi K(h_2^2 - h_1^2)}{\ln(r_2/r_1)}$	Dupuit's Eq.
Unsteady, Confined	$s = \frac{Q}{4\pi T} W(u)$	Theis Eq.
Unsteady (Approx)	$s = \frac{2.303Q}{4\pi T} \log\left(\frac{2.25Tt}{r^2 S}\right)$	Cooper-Jacob
Specific Capacity	$S_c = Q/s_w$	Well productivity
Well Loss	$s_w = BQ + CQ^2$	Jacob's Eq.

12 Past Exam Questions Review

Past Exam Question (2081 Final Assessment)

What are the assumptions of Theis Method? [4 Marks]

Past Exam Question (2078 Chaitra)

Define the terms cone of depression, radius of influence, and specific capacity. [6 Marks]

Past Exam Question (2073 Bhadra)

Derive an Expression for equation of steady radial flow into a well in confined aquifer with neat sketch. [8 Marks]

Past Exam Question (2071 Bhadra)

A well penetrates fully a confined aquifer 20m thick. When pumped at a rate of 2000 lpm, the drawdowns at 15m and 60m are 2m and 0.5m respectively. Calculate T and K.